

Practice Test

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1. Area Calculation (10 points)

Find the area of the bounded region in the first quadrant cut off by

$$3 = (\sqrt{x} + \sqrt{y})^2.$$

Answer:

$$\begin{aligned} A &= \int_0^3 (\sqrt{3} - \sqrt{x})^2 dx \\ &= \left[3x - 2\sqrt{2} \left(\frac{2}{3} \right) x^{3/2} + x^2/2 \right]_0^3 \\ &= 9 - 4\sqrt{6} + 9/2 \\ &= \frac{27 - 4\sqrt{6}}{2} \end{aligned}$$

□

2. Work Done by Pump (10 points)

A pump rated at 1000 ft-lbs per second is used to pump 250 cubic feet of water (weighing 62.4 lbs per cubic foot) from a reservoir at level $z = 0$ into a cylindrical tank with bottom 50 feet above the reservoir and radius 5 ft. Calculate how long this should take.

Answer: Notice that the tank will be filled to a height

$$H = \frac{250}{25\pi} = \frac{10}{\pi}.$$

The work required to pump the slice of water which ends up at height h in the tank is approximately

$$(50 + h)(62.4)(25\pi) \Delta h.$$

Thus, the total work required is

$$W = \int_0^{\pi/10} (50 + h)(62.4)(25\pi) dh = 25(62.4)\pi \left(5\pi + \frac{\pi^2}{200} \right).$$

This would be given in ft-lbs, and if we divide by the rating of the pump, we should obtain the total time in seconds:

$$\frac{62.4\pi^2(125 + \pi/8)}{1000} = \frac{15.6\pi^2(500 + \pi/2)}{1000} = \frac{7.8\pi^2(1000 + \pi)}{1000} = 7.8\pi^2 \left(1 + \frac{\pi}{1000} \right),$$

or about 77 seconds, or a little over a minute.

□

3. **Definite Integral** (10 points)

Compute the definite integral:

$$\int_{-2}^0 5^{-\theta} d\theta.$$

Answer: By the u-substitution $u = -\theta \ln 5$,

$$\begin{aligned}\int_{-2}^0 e^{-\theta \ln 5} d\theta &= -\frac{1}{\ln 5} \int_{\ln 25}^0 e^u du \\&= \frac{1}{\ln 5} e^u \Big|_{\ln 25}^0 \\&= \frac{1}{\ln 5} (e^0 - e^{\ln 25}) \\&= \frac{1}{\ln 5} (1 - 25) \\&= \frac{-24}{\ln 5}.\end{aligned}$$

□

4. **Initial Value Problem** (10 points)

Solve the initial value problem:

$$\begin{cases} \sqrt{x^2 - 9} y' = 1, & \text{for } x > 3, \\ y(5) = \ln 3. \end{cases}$$

Answer: Integrating $y' = \frac{1}{\sqrt{x^2 - 9}}$, we obtain

$$\begin{aligned}y(x) - y(5) &= \int_5^x \frac{1}{\sqrt{t^2 - 9}} dt \\&= \frac{1}{3} \int_5^x \frac{1}{\sqrt{(t/3)^2 - 1}} dt \\&= \int_{5/3}^{x/3} \frac{1}{\sqrt{u^2 - 1}} du.\end{aligned}$$

At this point, we make a trigonometric substitution $\sec \theta = u$, according to which $\cot \theta = \frac{1}{\sqrt{u^2 - 1}}$ and $du = \sec \theta \tan \theta d\theta$.

$$\begin{aligned}y(x) &= y(5) + \int_{5/3}^{x/3} \frac{1}{\sqrt{u^2 - 1}} du \\&= y(5) + \int_{\sec^{-1}(5/3)}^{\sec^{-1}(x/3)} \sec \theta d\theta \\&= y(5) + \ln(\sec \theta + \tan \theta) \Big|_{\sec^{-1}(5/3)}^{\sec^{-1}(x/3)} \\&= \ln(3) + \ln\left(\frac{x}{3} + \frac{\sqrt{x^2 - 9}}{3}\right) - \ln(3) \\&= \ln\left(\frac{x + \sqrt{x^2 - 9}}{3}\right).\end{aligned}$$

□

5. **First-Order Linear Equation** (15 points)

Solve the first-order linear equation:

$$xy' + 3y = \frac{\sin x}{x^2}, \quad \text{for } x > 0.$$

Answer: Putting the equation into standard form, we get

$$y' + \frac{3}{x}y = \frac{\sin x}{x^3}.$$

Thus, the integrating factor should be

$$\mu = e^{\int (3/x) dx} = e^{3 \ln x} = x^3.$$

Multiplying the equation by the integrating factor, we get

$$x^3 y' + 3x^2 y = (x^3 y)' = \sin x$$

Integrating, from some $x_0 > 0$ to x ,

$$x^3 y - x_0^3 y(x_0) = -\cos x + \cos x_0$$

Therefore,

$$y = \frac{1}{x^3} [-\cos x + \cos x_0 + x_0^3 y(x_0)]$$

In this expression x_0 can be any positive constant, and $y(x_0)$ can be taken to be any constant. Alternatively, we can write simply

$$y = \frac{1}{x^3} [-\cos x + C]$$

where C is an arbitrary constant.

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6. **Sequence Convergence** (15 points)

Discuss the convergence of the sequence:

$$\frac{(-4)^n}{n!}.$$

Answer: The sequence converges to 0 as $n \rightarrow \infty$, since $\frac{(-4)^n}{n!} \rightarrow 0$ due to the factorial in the denominator growing faster than the exponential in the numerator.

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7. **Series Convergence** (15 points)

Discuss the convergence of the series

$$\sum_{j=1}^{\infty} \frac{(-1)^{j+1}(x+2)^j}{j2^j}$$

Answer: Denoting the j -th term by a_j , we attempt to use the ratio test:

$$\lim_{j \rightarrow \infty} \frac{|a_{j+1}|}{|a_j|} = \lim_{j \rightarrow \infty} \frac{|(x+2)|j2^j}{(j+1)2^{j+1}} = \frac{|x+2|}{2}.$$

Thus, the series converges absolutely for $|x+2| < 2$ and diverges for $|x+2| > 2$. When $x = -4$, the series becomes

$$\sum \frac{-1}{j}$$

This is the negative harmonic series and diverges, for example, by the integral test. When $x = 0$, the series becomes

$$\sum -\frac{(-1)^j}{j}$$

This is the (negative) alternating harmonic series and converges by the alternating series test.

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8. **Taylor Series Expansion** (15 points)

Find the Taylor series expansion for $\cos(5x^2)$ at $x = 0$ and discuss the convergence of the series.

Answer: We know

$$\cos x = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j)!} x^{2j}$$

Therefore,

$$\cos(5x^2) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j)!} x^{4j}$$

and this series converges (absolutely) for all x since the series for $\cos x$ converges for all x .

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